

AMIT GAUTAM

Senior Backend Engineer • Distributed Systems & AI Integration • Java & Python • 7+ Years

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Braunschweig, Germany • Open to Relocation: EU • UK • AU • NZ • Remote

SUMMARY

Backend engineer with 7+ years building distributed systems in Java 17-21/Spring Boot and Python/FastAPI. Sole backend developer at Smart Microwave Sensors in a full-stack role, and the team's final reviewer for all backend and frontend code and design decisions. Delivered 20+ production microservices at Walmart, built event-driven pipelines with Kafka, ran services on Kubernetes, and owned workflow automation with Camunda 7. Built codit, an AI editing agent that runs inside DaVinci Resolve, using the Claude Agent SDK and FastMCP, including reverse-engineering Resolve's timeline and audio formats. Strong on system design, security (JWT/OAuth2), observability, and CI/CD. Looking for senior backend or AI-backend roles. Open to relocation across EU, UK, and Australia/NZ.

TECHNICAL SKILLS

Languages	Java 17-21 • Python 3.x • JavaScript (ES6+) • TypeScript • Shell Scripting
Backend	Spring Boot • FastAPI • Hibernate • REST APIs • GraphQL • Microservices
AI & LLM	LLM Integration • RAG • LangChain • LangGraph • FastMCP (MCP) • Claude Agent SDK • pgvector • Qdrant • Tree-sitter (AST) • Prompt Engineering • AI Agents • LangSmith • Groq • OpenAI Embeddings
Cloud & DevOps	GCP (BigQuery, GCS, Pub/Sub) • Kubernetes (Helm) • Docker • Docker Compose • GitHub Actions • GitLab CI • Maven • Gradle • Linux/Ubuntu • Nginx • Railway • Vercel
Messaging	Apache Kafka • MQTT • Event-Driven Architecture • Async Processing • Celery • Redis
Security	JWT • OAuth2 • Sidecar-based Service Auth • Spring Security
Workflow/BPM	Camunda 7 (Certified) • BPMN 2.0 (Certified) • Process Automation
Databases	PostgreSQL • pgvector • MongoDB • Redis • BigQuery • SQL Server
Observability	Grafana • Prometheus • LangSmith • Distributed Tracing • SLO/SLA • Incident Response
Practices	System Design • DDD • Agile/Scrum • TDD • Mentoring • Technical Leadership • Code Review • On-Call • OpenAPI/Swagger • Postman • Mockoon

WORK EXPERIENCE

Senior Full Stack Engineer (Backend-Focused) — Smart Microwave Sensors GmbH • Braunschweig, Germany Apr 2025 – Present

- Sole backend developer on the team in a full-stack role; own backend architecture and API design, and handle major frontend development alongside the frontend engineers.
- Own PR review and final architecture/API design decisions for the backend; onboarded and mentored a new hire on the backend stack and workflows.
- Final reviewer for all frontend code and design changes, including sign-offs requested by the team lead; mentored two frontend engineers on code quality and design.
- Own API contract design; gather requirements from downstream teams in daily syncs and translate them into REST API contracts before implementation.
- Coordinate with the embedded Linux team on Yocto-level changes for solutions that can't run directly on-device due to security or platform constraints.
- Built a browser-based MQTT traffic analysis tool (Python, React) extending MQTT Explorer with configurable-frequency message polling for real-time traffic analysis.
- Built COMHUB, the real-time traffic monitoring system, solo from a blank repo to production: FastAPI backend with MQTT ingestion from 100+ road intersections, REST APIs to a React dashboard, sub-200ms latency via MQTT broker tuning.
- Owned the graph component service end to end: requirements gathering, POC, and the Recharts dashboard showing live per-intersection traffic data.
- Built the D3.js/SVG map layer on Google Maps and Bing Maps for real-time sensor positions, lane markings, and detection zones.
- Re-architected Mockoon so only explicitly listed APIs return mocked data (default behavior wasn't compatible with the existing frontend architecture), cutting frontend blocking time by roughly 2 sprints per feature cycle.
- Follow TDD in the FastAPI backend, writing unit and integration tests with pytest; regression tests run as part of the CI pipeline before every deploy.
- Containerized services with Docker; set up a CI pipeline for automated tests and deployment on every merge to main.

Software Engineer 3 – Backend Platform — Walmart Global Tech • Bengaluru, India Jan 2022 – Mar 2025

- Owned 20+ Java 17/Spring Boot microservices across supply chain, pricing, and promotions, serving millions of daily requests on Walmart.com.
- Queried pricing data from the pricing team's GraphQL API as part of microservice integrations.

- Built Wrangler, a Java/Camunda 7 wrapper adopted by 10+ teams; reduced workflow integration time from 2-3 weeks to a few days.
- Built Execution Services, a Java PaaS between Camunda and external task processors, adopted by 10+ downstream teams for polling, execution, callbacks, and retry logic.
- Replaced synchronous HTTP calls across 15-20+ service endpoints with Kafka-based async messaging, removing the main throughput bottleneck.
- Implemented JWT/OAuth2-based authentication and sidecar containers for upstream/downstream service-to-service auth, following established platform security patterns.
- Set up GitHub Actions across 20+ services for automated tests, quality gates, and blue-green deploys; built Grafana/Prometheus dashboards that cut MTTD on P1 incidents by around 40%.
- Practiced TDD across Java/Spring Boot microservices, writing unit and integration tests with JUnit; regression suites ran as GitHub Actions quality gates before every deploy.
- Deployed and operated services on Kubernetes: wrote Helm charts, scaled pods up and down based on load, and debugged production issues like OOM kills and pod restarts via logs.
- Built Grafana dashboards on Kubernetes/Prometheus infra metrics tracking error counts and response counts across services.
- Authored API/backend and Grafana dashboard documentation (Prometheus and SQL query data) on Confluence, used as reference material by other engineering teams.
- Built Cost Driver, a hackathon project forecasting grocery item prices from ingredient costs to maximize profit margins; presented at Walmart's hackathon showcase.
- Wrote BigQuery queries to pull retail forecast data from the data science team's datasets and fed workflow-generated data back for their forecasting models.
- Used GCP Cloud Storage for storing and retrieving workflow-related artifacts in the Camunda-based services.
- 3 years on the senior on-call rotation; resolved or led resolution on dozens of P1 incidents and authored the domain runbooks. Multiple Impact Driver and Firefighter awards for P1 resolution speed.

Teaching Assistant – Algorithms (Part-Time) — **Coding Ninjas** • Remote Aug 2021 – Dec 2021

- Taught DSA in Python to 100+ students and working professionals. Received a 4.9/5 rating.

Software Engineering Specialist – Backend — **Dassault Systèmes** • Pune, India Jan 2020 – Jan 2022

- Owned the CRS Framework REST APIs, used in production by Ericsson, Melexis, and Framatome.
- Developed TRM, replacing IBM DOORS for requirements traceability across PLM lifecycle data; set up the team's first GitLab CI pipelines.

Associate IT Developer — **India Medtronic Pvt. Ltd.** • Bengaluru, India Feb 2019 – Jan 2020

- Built NCG, a Python/Flask tool auto-generating Cisco Switch 3850-48 configs from templates, replacing a manual process across Medtronic's global network infrastructure.

KEY PROJECTS

codit – AI Video-Editing Agent for DaVinci Resolve | *Personal project* | github.com/iamamit/codit (private)

- **codit (agent)**: chat panel inside DaVinci Resolve (Electron + a Claude Agent SDK server) that cuts, grades, mixes and fixes audio from plain-language requests, checks its own result by measuring and rendering the audio, and undoes changes without losing the user's manual edits.
- **codit-core (MCP server)**: FastMCP engine that gives the agent what Resolve's API can't. I reverse-engineered Resolve's timeline and Fairlight formats (protobuf/zstd blobs, a zlib object store) with a UI crawler I wrote (macOS input events + Vision OCR) and byte diffing, mapping the mixer, automation and 20+ effects; plus speech-aware loudness with FFmpeg (EBU R128).
- **Video template engine**: reusable motion-graphics templates rendered frame-exact (headless Chrome → FFmpeg, transparent video) with search-ready metadata, so the agent fills a template instead of writing animation code: fewer tokens and one consistent style. The goal: an AI-first video editor of my own.

Stack: Python · Claude Agent SDK · FastMCP (MCP) · Electron · FFmpeg · headless Chrome · Protobuf · pytest

EDUCATION

B.Tech – Computer Science & Engineering — **Harcourt Butler Technical University** • Kanpur, India 2013 – 2017

CERTIFICATIONS & RECOGNITION

- Camunda 7 Java Developer Certified · BPMN 2.0 Certified · Google Cloud Digital Leader
- 15 Walmart badges · Multiple Impact Driver & Firefighter awards for P1 resolution
- Certificate of Excellence – DSA in Python, Coding Ninjas (Top 10%)

LANGUAGES

English (Fluent) · Hindi (Native) · German (Learning – A1)